

# **Migrating SQL Server to Azure**

Exhaustive inventory of targets, methods & tools | v1.2

July 2026

## Contents

|                                                                                 |           |
|---------------------------------------------------------------------------------|-----------|
| <b>1 Why migrate in 2026 (the short “why now”)</b>                              | <b>2</b>  |
| <b>2 Taxonomy — separate targets, control planes, and methods</b>               | <b>3</b>  |
| <b>3 The Azure targets (8 families)</b>                                         | <b>4</b>  |
| <b>4 Control planes &amp; assessment experiences</b>                            | <b>5</b>  |
| 4.1 Microsoft tooling by source → target (assess · data · schema) . . . . .     | 5         |
| <b>5 Migration methods per target</b>                                           | <b>5</b>  |
| 5.1 To SQL Server on Azure VM (IaaS rehost) . . . . .                           | 6         |
| 5.2 To Azure SQL Managed Instance (managed lift-and-shift) . . . . .            | 6         |
| 5.3 To Azure SQL Database (cloud-native refactor) . . . . .                     | 6         |
| 5.4 To SQL database in Fabric ( <i>Preview</i> ) . . . . .                      | 7         |
| 5.5 To Containers / Arc-enabled SQL MI . . . . .                                | 7         |
| 5.6 Offline & network transfer accelerators (large estates) . . . . .           | 7         |
| <b>6 Ancillary components (the cutover blockers)</b>                            | <b>8</b>  |
| <b>7 Downtime strategy (cutover window) — the #1 architect criterion</b>        | <b>8</b>  |
| <b>8 Summary matrix — method / tool × target</b>                                | <b>9</b>  |
| <b>9 Source-version &amp; retirement reference</b>                              | <b>9</b>  |
| <b>10 Cross-cloud sources &amp; reverse migration</b>                           | <b>10</b> |
| <b>11 Third-party alternatives (when they beat the native stack)</b>            | <b>10</b> |
| <b>12 Decision criteria &amp; “when to recommend what”</b>                      | <b>11</b> |
| <b>13 Field insights — recurring pitfalls</b>                                   | <b>11</b> |
| <b>14 FY27 SQL Motion context &amp; AI Migration Agent</b>                      | <b>12</b> |
| <b>15 Commercial levers &amp; funding programs (FY27)</b>                       | <b>13</b> |
| 15.1 Commercial / licensing levers (permanent TCO reducers) . . . . .           | 13        |
| 15.2 Microsoft-funded engagement programs . . . . .                             | 14        |
| 15.3 Partner mechanics . . . . .                                                | 14        |
| <b>16 Sources (Microsoft Learn)</b>                                             | <b>14</b> |
| <b>17 Document version &amp; changelog</b>                                      | <b>16</b> |
| <b>18 Appendix — Using the sql-migration-advisor Copilot CLI skill</b>          | <b>17</b> |
| 18.1 Install (once) . . . . .                                                   | 17        |
| 18.2 Example output . . . . .                                                   | 17        |
| <b>Goal.</b> Exhaustively list every way and every tool to migrate a SQL Server |           |

database to an Azure service (all PaaS, including containers) or a VM (Azure VM / Azure VMware Solution).

**Audience.** Partners, architects, and customer DBAs — usable in pre-sales and as the knowledge base behind the *SQL in a Day* AI Migration Agent (§14).

**Verification.** Tool retirements, version requirements and target families were cross-checked against Microsoft Learn and product announcements (current as of July 2026). Links are gathered in §16 Sources.

**Version.** v1.2 — July 2026. Change history in §17 Document version & changelog.

**IMPORTANT 2025-2026 tooling reset — read this first.** - **Data Migration Assistant (DMA)** is RETIRED (16 July 2025). - **Azure Data Studio (ADS)** and its Azure SQL Migration extension are RETIRED (28 February 2026). - The new entry point is the **SSMS 22 Migration Component** (assess + migrate from SSMS), complemented by SQL Server migration in Azure Arc (portal, with Copilot) and Azure Database Migration Service (DMS). - Modern **DMS** supports offline-only migration to Azure SQL Database (online / minimal-downtime is available for Managed Instance and SQL VM targets, per the [DMS supported-scenarios](#) matrix). - **Azure DMS classic** SQL Server scenarios are RETIRED (15 March 2026) — use the modern DMS (an Azure resource, via portal / PowerShell / CLI).

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## 1 Why migrate in 2026 (the short “why now”)

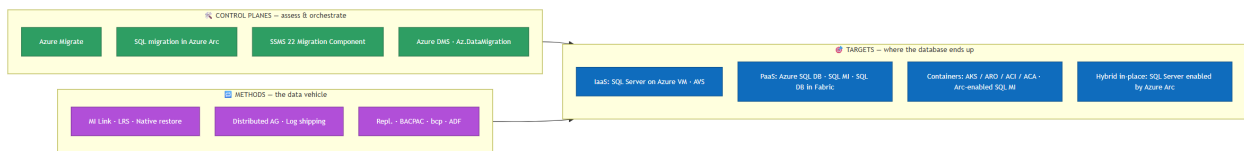
- **AI runs on data.** Modern, managed databases are the foundation; SQL Server 2025 / Azure SQL add a native vector type + functions, **DiskANN** vector indexing (high-throughput ANN search), native json, and zero-ETL **Fabric Mirroring** (analytics on OneLake with no pipelines) — making the estate AI-ready and improving migration (enhanced distributed AG).
- **End-of-support pressure.** Out-of-support SQL Server/Windows versions push modernization; Extended Security Updates (ESU) are free on Azure VMs, and for on-prem are delivered only via Azure Arc (paid) — this changes the *stay vs migrate* math.
- **Cost levers.** Azure Hybrid Benefit (AHB) applies to SQL DB (vCore), Managed Instance and VM (not to Fabric SQL DB) — up to ~85% savings when combined with reservations/ESU.
- **Partner leverage.** Sharing a deal with a partner increases win rate and deal size; programs like Cloud Accelerate Factory and SQL in a Day industrialize delivery (§14). Commercial & funding levers are detailed in §15.
- **Concrete forcing function.** SQL Server 2016 reaches End of Support on 14 July 2026 — the #1 trigger in EMEA. “Stay” then means paying ESU; “migrate” makes ESU free on Azure.

| Version         | Extended End of Support          | Status (Jul 2026)           |
|-----------------|----------------------------------|-----------------------------|
| SQL Server 2012 | 12 Jul 2022 — ESU ended Jul 2025 | ✗ Out of support            |
| SQL Server 2014 | 9 Jul 2024 — ESU ~Jul 2027       | Out of support (ESU window) |
| SQL Server 2016 | 14 Jul 2026 — ESU ~Jul 2029      | EOS imminent                |
| SQL Server 2017 | 12 Oct 2027                      | ✓ Extended support          |
| SQL Server 2019 | 8 Jan 2030                       | ✓                           |
| SQL Server 2022 | 11 Jan 2033                      | ✓                           |
| SQL Server 2025 | GA 18 Nov 2025                   | ✓ Latest                    |

Windows Server often sits under SQL Server — **WS 2012/2012 R2** is out of support (ESU to Oct 2026), WS 2016 EOS ~Jan 2027. Factor the OS into the move.

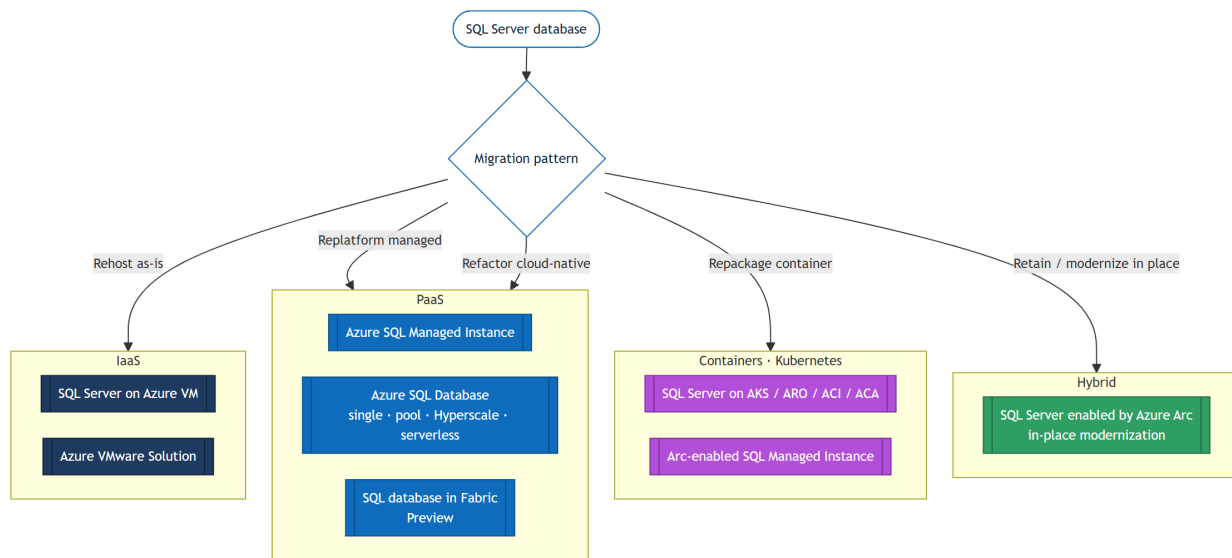
## 2 Taxonomy — separate targets, control planes, and methods

The single most common mistake is to mix these three layers. Keep them distinct:



- **Targets** = runtime destinations (where the DB lives).
- **Control planes / experiences** = how you assess, recommend and orchestrate (Azure Migrate, Arc, SSMS 22, DMS). Azure Arc-enabled SQL Server is a control plane, not a runtime target.
- **Methods** = the actual data-movement vehicle (MI Link, LRS, backup/restore, replication...).

### 3 The Azure targets (8 families)



| # | Target                                           | Layer                  | When to choose it                                                                                                                                                                                                                                                                                                                                                                            | Compatibility                                                                                        | Doc                                 |
|---|--------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------|
| 1 | SQL Server on Azure VM                           | IaaS                   | Faithful lift & shift: OS / file-system control, exact version, FileStream/FileTable, PolyBase, cross-instance DTC, third-party agents.                                                                                                                                                                                                                                                      | Full                                                                                                 | <a href="#">overview</a>            |
| 2 | Azure VMware Solution (AVS)                      | IaaS                   | Zero-refactor data-center exit for existing VMware estates; keeps FCI and Always On AG; migrate with VMware HCX / vMotion.                                                                                                                                                                                                                                                                   | Full                                                                                                 | <a href="#">AVS</a>                 |
| 3 | Azure SQL Managed Instance                       | PaaS                   | Managed lift-and-shift: keep instance objects (logins, SQL Agent, server triggers, cross-DB, linked servers), native vNet. Tiers GP / BC / Next-gen GP (GA since Nov 2025 — Elastic SAN backend: 500 DBs, 128 vCores, 32 TB, 80K IOPS, 192 MB/s log, 3–4 ms latency, configurable IOPS/memory; ~5x better price-per-DB by density). Free offer: 1 instance / 12 months. No Hyperscale on MI. | ~Near-full (instance)                                                                                | <a href="#">overview</a>            |
| 4 | Azure SQL Database                               | PaaS                   | Cloud-native app / microservice. Models: single DB / elastic pool; tiers GP / BC / Hyperscale; purchasing vCore / DTU / serverless. Hyperscale scales to 128 TB (large / HTAP); serverless for intermittent; elastic pools for consolidation. Free offer: 10 serverless DBs for the subscription lifetime.                                                                                   | Database surface (no instance-level)                                                                 | <a href="#">overview</a>            |
| 5 | SQL database in Fabric (Preview for migration)   | PaaS                   | Fabric-native OLTP unified with OneLake. Migrate via Fabric Migration Assistant (DACPAC schema ≤ 20 MB, on-prem data gateway only, no Private Link). Not an enterprise OLTP target yet.                                                                                                                                                                                                      | Subset; preview                                                                                      | <a href="#">Migration Assistant</a> |
| 6 | SQL Server in containers — AKS / ARO / ACI / ACA | Container              | Full control of the engine in a container (dev/test, edge, custom). Pod + PersistentVolume; HA via the Kubernetes scheduler.                                                                                                                                                                                                                                                                 | High — SQL on Linux (no FILESTREAM/FileTable, SSRS/SSAS/SSIS, ML Services; SQL Agent off by default) | <a href="#">SQL on Kubernetes</a>   |
| 7 | Azure Arc-enabled SQL Managed Instance           | Container (PaaS)       | Managed SQL MI engine on any Kubernetes (AKS, ARO, EKS, GKE, OpenShift) via kubectl + CRD. Sovereignty / edge / multi-cloud.                                                                                                                                                                                                                                                                 | ~Same as SQL MI                                                                                      | <a href="#">create Arc SQL MI</a>   |
| 8 | SQL Server enabled by Azure Arc                  | Hybrid (control plane) | Not a runtime target — the bootstrap pillar: connect on-prem SQL <i>without touching it</i> → weekly continuous assessment, blocker detection, ESU, PAYG (OpEx) licensing, governance, and a portal Copilot-assisted Database migration to MI/VM (MI Link / LRS). Sources SQL 2012+.                                                                                                         | n/a                                                                                                  | <a href="#">Arc migration</a>       |

**Lift & shift is not VM-only.** Both SQL Server on Azure VM (IaaS) and Azure SQL Managed Instance (managed PaaS) are valid lift-and-shift targets. Since 2024, MI compatibility (SQL Agent, cross-DB queries, linked servers...) makes it the default *managed* lift-and-shift.

## 4 Control planes & assessment experiences

| Tool / experience                                          | Role                                                                                               | Status (2026)                                 | Notes                                                                                                                                                                       |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Azure Migrate                                              | Discovery / assessment / sizing / business case at scale                                           | GA (+ Arc-based agentless discovery, Preview) | Appliance (VMware/Hyper-V/Physical) or import-based or Arc-based. Right-sizes SQL DB / MI / VM.                                                                             |
| SQL Server migration in Azure Arc                          | Portal-driven assess + migrate for any Arc-enabled SQL Server                                      | GA; MI target GA, VM target Preview           | Copilot recommends method (MI Link vs LRS); continuous assessment; supports sources from SQL Server 2012+.                                                                  |
| SSMS 22 Migration Component                                | DBA-first entry point: assess + launch the right path from SSMS                                    | GA (Windows-only)                             | Replaces DMA / ADS extension. Backup/restore, MI Link, DMS. Azure SQL assessment capability expected ~Q3 CY2026 ( <i>public roadmap, subject to change</i> ).               |
| Azure DMS (modern)                                         | Managed migration orchestration (Azure resource · portal / PowerShell / CLI)                       | GA                                            | Use the modern DMS — DMS <i>classic</i> SQL scenarios retired 15 Mar 2026. Offline-only to Azure SQL DB; online/minimal-downtime to MI / SQL VM (MI Link preferred for MI). |
| PowerShell<br><a href="#">Az.DataMigration</a> / Azure CLI | Automate DMS at scale (CI/CD)                                                                      | GA                                            | Often the only viable path beyond ~50 databases.                                                                                                                            |
| SSMA                                                       | Heterogeneous conversion (schema/code/data)                                                        | GA                                            | Oracle / Sybase / DB2 / MySQL / Access → Azure SQL. Not for homogeneous SQL→SQL.                                                                                            |
| Database Experimentation Assistant (DEA)                   | Capture + replay a production workload on the target to validate performance <i>before</i> cutover | Retired 15 Dec 2024                           | Replaced by the open-source SQLWorkload / WorkloadTools. The <i>concept</i> (catch plan/compat regressions before cutover) remains valuable; the MS tool itself is retired. |

**NOTE Retired — do not use in new runbooks:** DMA (16 Jul 2025) and Azure Data Studio + Azure SQL Migration extension (28 Feb 2026). For cross-platform SQL dev, use VS Code + MSSQL extension; migration work continues via SSMS 22 / Azure Arc / DMS / [Az.DataMigration](#) CLI. SQL Data Sync retires 30 Sep 2027 — don't build new sync/migration on it (use ADF, transactional replication or AG).

### 4.1 Microsoft tooling by source → target (assess · data · schema)

The Microsoft-recommended tool per source/target combination (homogeneous and heterogeneous):

| Source                   | Target            | Assess                     | Data migration                 | Schema          |
|--------------------------|-------------------|----------------------------|--------------------------------|-----------------|
| SQL Server (Arc-enabled) | Azure SQL MI      | SQL migration in Azure Arc | SQL migration in Azure Arc     | Not needed      |
| SQL Server (not Arc)     | Azure SQL VM / MI | Azure Migrate              | DMS                            | Not needed      |
| SQL Server               | Azure SQL DB      | Azure Migrate              | STRIM (online) · DMS (offline) | DMS             |
| Sybase                   | Azure SQL         | Azure Migrate              | STRIM                          | SSMA for Sybase |
| Oracle                   | Azure SQL         | Azure Migrate              | STRIM                          | SSMA for Oracle |

GitHub Copilot is applicable as AI-assisted migration tooling (schema/code conversion inside SSMA; app-level via GitHub Copilot App Modernization for .NET / Java). Portfolio & application/code assessment partners complementary to Azure Migrate: Dr Migrate, CAST Highlight, UnifyCloud.

## 5 Migration methods per target

Standardized columns (Microsoft Learn style): **Method · Min source · Target/min · Downtime · Key constraints.**

## 5.1 To SQL Server on Azure VM (IaaS rehost)

| Method                                                    | Min source   | Downtime             | Key constraints / notes                                                              |
|-----------------------------------------------------------|--------------|----------------------|--------------------------------------------------------------------------------------|
| Azure Migrate (lift & shift)                              | SQL 2008 SP4 | Online (replication) | Whole VM/instance, incl. FCI and AG; up to ~35,000 VMs.                              |
| Distributed availability group (DAG)                      | SQL 2016     | Near-zero            | Reuse on-prem AG; needs AD Domain Services (or workgroup AG + certs) and ports open. |
| Backup to a file (.bak) + copy                            | SQL 2008 SP4 | Offline              | Simple, supports > 1 TB; use compression / multi-file split for WAN.                 |
| Backup to URL (Azure Blob)                                | SQL 2014     | Offline              | 12.8 TB (SQL 2016+) / 1 TB otherwise; for > 1 TB use local backup + AzCopy.          |
| Detach & attach (MDF/LDF via Blob)                        | SQL 2008     | Offline              | For very large DBs where backup/restore is too slow.                                 |
| Log shipping                                              | SQL 2008     | Minimal              | Windows-only (not for SQL on Linux sources).                                         |
| Always On AG                                              | SQL 2012     | Near-zero            | Fail an existing AG onto Azure VM replicas.                                          |
| Convert machine to VHD / Ship hard drive / Azure Data Box | any          | Offline              | Estate exit with limited WAN; multi-TB .bak / .bacpac via Data Box.                  |

## 5.2 To Azure SQL Managed Instance (managed lift-and-shift)

| Method                          | Min source                                                           | Downtime           | Key constraints / notes                                                                                                                                                                                                                                    |
|---------------------------------|----------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Managed Instance link (MI Link) | SQL 2016 and later (incl. 2022, 2025; Win Server 2016+, Ent/Std/Dev) | Near-zero (online) | Distributed-AG based; R/O readable target during migration; reverse fallback to SQL 2022 / 2025 (DR / Azure exit); up to 10 simultaneous DBs (Arc ext 1.1.3348.364+); needs port 5022 both ways.                                                           |
| Log Replay Service (LRS)        | SQL 2012 (all editions)                                              | Offline (planned)  | Full/diff/log → Azure Blob; public endpoint; 30-day max window; FULL recovery + NORECOVERY (no reads); the only managed path that applies differentials to MI; GP & BC supported, but for large Business Critical migrations prefer MI Link (true online). |
| Native backup & restore (.bak)  | SQL 2008                                                             | Offline            | Simplest; migrate TDE certificate <i>before</i> restore or it fails late; master/msdb restore not supported (script instance objects).                                                                                                                     |
| Transactional replication       | SQL 2012                                                             | Online             | Replicate all/part; article-type limits.                                                                                                                                                                                                                   |
| bcp / Smart Bulk Copy           | any                                                                  | Offline            | High-speed data-only / partial (parallel copy).                                                                                                                                                                                                            |
| BACPAC / SqlPackage             | any                                                                  | Offline            | Smaller DBs / simple.                                                                                                                                                                                                                                      |
| Azure Data Factory — Copy       | any                                                                  | Offline / batch    | When migration = integration / transformation.                                                                                                                                                                                                             |

## 5.3 To Azure SQL Database (cloud-native refactor)

| Method                    | Min source         | Downtime           | Key constraints / notes                                                                                                                          |
|---------------------------|--------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Azure DMS (offline)       | SQL 2008+          | Offline            | Offline only to Azure SQL DB (online/minimal-downtime is available for MI / SQL VM, not SQL DB).                                                 |
| STRIM (online CDC)        | any                | Online (near-zero) | Microsoft-recommended online / CDC data migration to Azure SQL DB — fills the gap that DMS is offline-only for SQL DB; pair with DMS for schema. |
| Transactional replication | SQL 2016–2019 only | Online             | Push subscription only; subset of tables/columns/rows; article-type limits (no <code>hierarchyid</code> , <code>sql_variant</code> ...).         |
| BACPAC / SqlPackage       | any                | Offline            | Small/medium; SqlPackage for scale.                                                                                                              |
| bcp / Smart Bulk Copy     | any                | Offline            | Data-only / bulk.                                                                                                                                |
| Azure Data Factory — Copy | any                | Offline / batch    | BI / integration.                                                                                                                                |

**X Not supported to Azure SQL Database:** native .bak restore, detach/attach, MI Link. SQL Agent → use [Elastic Jobs](#).

**DACPAC vs BACPAC.** A DACPAC packages the schema only; a BACPAC packages schema + data. Both are produced/consumed by SqlPackage and the

VS Code MSSQL extension — a simple way to migrate (BACPAC) or version (DACPAC) offline.

## 5.4 To SQL database in Fabric (*Preview*)

| Method                              | Downtime         | Key constraints / notes                                                                                                                                                          |
|-------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fabric Migration Assistant — DACPAC | Offline          | Schema via DACPAC ≤ 20 MB; AI-assisted compatibility fixes; data via Fabric Data Factory copy job + on-prem data gateway (no VNet gateway / Private Link).                       |
| Fabric Mirroring for SQL Server     | n/a (continuous) | NOT a one-shot migration — near-real-time CDC replication to OneLake for analytics. GA (Nov 2025), optimized for SQL Server 2025. Complementary "analytical modernization" path. |

## 5.5 To Containers / Arc-enabled SQL MI

| Target                           | Method                                                             | Downtime                 | Notes                                                                                      |
|----------------------------------|--------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------|
| Arc-enabled SQL MI (AKS/ARO/...) | Native backup/restore, point-in-time restore                       | Offline                  | Requires Arc data controller; exposes a SQL MI endpoint → logical vehicles of \$5.2 apply. |
| SQL Server container (mcr image) | Backup/Restore via mounted volume, detach/attach, BACPAC, bcp, ADF | Offline / Online (repl.) | Persist on Azure Disk (AKS/ARO) / Azure Files (ACI/ACA); HA via scheduler.                 |

**WARNING Containers are not a fully-managed cloud database.** Plain mssql containers (AKS / ARO / ACI / ACA) put patching, high availability and backups entirely on you — best suited to dev/test and edge scenarios. Arc-enabled SQL MI automates engine patching, backups and HA through the Arc data controller, but you still own the Kubernetes cluster, persistent storage and DR. Neither is equivalent to the fully-managed Azure SQL Database / Managed Instance PaaS.

## 5.6 Offline & network transfer accelerators (large estates)

For multi-terabyte databases the network — not the method — is usually the bottleneck. Plan the **bulk transfer / initial seed** explicitly:

| Accelerator                            | Use it for                                                            | Notes                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Azure Data Box / Data Box Heavy        | One-shot physical move of multi-TB .bak / .bacpac or a whole VM fleet | Data Box ≈ 80 TB usable; Data Box Heavy ≈ 770 TB for hundreds of TB. Microsoft recommends Data Box to migrate a SQL server/estate and as the initial <i>seed</i> before syncing the delta over the network. |
| ExpressRoute / dedicated circuit       | Sustained high-throughput online transfer & seed                      | Avoids WAN time-outs on long-running backup/restore, LRS or replication.                                                                                                                                    |
| AzCopy + Backup-to-URL                 | Push large local backups to Blob                                      | For DBs > 1 TB: local backup + AzCopy is faster/safer than direct Backup-to-URL.                                                                                                                            |
| Azure Storage Mover                    | Orchestrated bulk file transfer on-prem → Azure Files / Blob          | Centralized, resumable; good for backup repositories.                                                                                                                                                       |
| Compressed / split / encrypted backups | Shrink & secure the payload in transit                                | Compression + multi-file split optimize WAN; backup encryption (SQL 2014+) for compliance during transit.                                                                                                   |

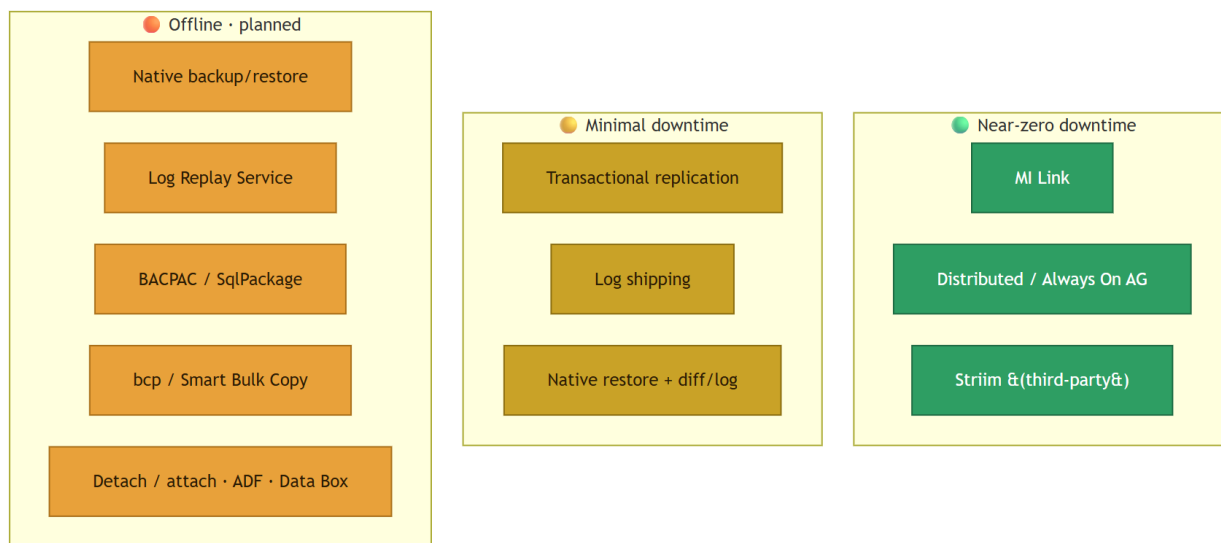
**Seed-then-sync pattern.** For very large or low-downtime migrations: ship the initial full backup via Data Box (or AzCopy over ExpressRoute), then catch up the delta with log restore (LRS), MI Link, transactional replication or log shipping before cutover.

## 6 Ancillary components (the cutover blockers)

Databases rarely move alone — these sub-components block go-live if forgotten:

| Component                 | Path to Azure                                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------|
| SQL Agent jobs            | MI: native · SQL DB: <a href="#">Elastic Jobs</a>                                                                     |
| Logins / users            | Script + recreate; DMS does not migrate Windows logins by default (enable option + grant MI read access to Entra ID). |
| SSIS                      | <a href="#">Azure-SSIS Integration Runtime</a> (SSISDB via DMS).                                                      |
| SSRS                      | <a href="#">Power BI paginated reports</a> (RDL).                                                                     |
| SSAS                      | Azure Analysis Services or Power BI Premium (XMLA).                                                                   |
| Linked servers / cross-DB | Supported on VM/MI; not on SQL DB — refactor required.                                                                |
| TDE                       | Migrate the server-level certificate <i>before</i> any native restore to MI.                                          |

## 7 Downtime strategy (cutover window) — the #1 architect criterion



**Sizing rule.** Never size MI/SQL DB on *average* CPU. Use a Perfmon baseline  $\geq 7$  days + ~20% headroom for cutover. Network: don't estimate size ÷ bandwidth — Backup-to-URL throughput is capped by the Blob layer; test with AzCopy + one full backup before committing a go-live date, and plan a rollback / fallback window in case the estimate proves optimistic.

## 8 Summary matrix — method / tool × target

| Method / tool              | SQL VM  | AVS | SQL MI     | SQL DB         | Fabric SQL DB | Arc SQL MI | SQL container |
|----------------------------|---------|-----|------------|----------------|---------------|------------|---------------|
| Azure Migrate (assess)     | ✓       | ✓   | ✓          | ✓              | –             | –          | –             |
| DMS                        | ✓       | –   | ✓ (online) | ✓ (offline)    | ✗             | –          | –             |
| MI Link                    | reverse | ✗   | ✓          | ✗              | ✗             | ✓          | ✗             |
| Log Replay Service         | ✗       | ✗   | ✓          | ✗              | ✗             | –          | ✗             |
| Native backup/restore      | ✓       | ✓   | ✓          | ✗              | ✗             | ✓          | ✓             |
| Distributed / Always On AG | ✓       | ✓   | ✗          | ✗              | ✗             | ✗          | ✗             |
| HCX / vMotion              | ✗       | ✓   | ✗          | ✗              | ✗             | ✗          | ✗             |
| Transactional replication  | ✓       | ✓   | ✓          | ✓ <sup>1</sup> | ✗             | ✓          | ✓             |
| BACPAC / SqlPackage        | ✓       | ✓   | ✓          | ✓              | ✓ (DACPAC)    | ✓          | ✓             |
| bcv / Smart Bulk Copy      | ✓       | ✓   | ✓          | ✓              | –             | ✓          | ✓             |
| ADF Copy                   | ✓       | ✓   | ✓          | ✓              | ✓             | ✓          | ✓             |
| Fabric Migration Assistant | ✗       | ✗   | ✗          | ✗              | ✓             | ✗          | ✗             |
| SSMA (heterogeneous)       | ✓       | ✓   | ✓          | ✓              | –             | ✓          | ✓             |

✓ supported · ✗ n/a · – indirect/non-first-class · reverse only · <sup>1</sup> SQL DB transactional replication: source **SQL 2016–2019** only.

## 9 Source-version & retirement reference

**Tooling timeline:** DMA retired Jul 2025 → ADS + SQL Migration extension retired Feb 2026 → DMS *classic* SQL scenarios retired Mar 2026 → SQL Data Sync retires Sep 2027.

| Item                                        | Status / requirement                                                            |
|---------------------------------------------|---------------------------------------------------------------------------------|
| DMA                                         | Retired 16 Jul 2025 → SSMS 22 / Arc / Azure Migrate                             |
| Azure Data Studio + SQL Migration extension | Retired 28 Feb 2026 → VS Code + MSSQL; SSMS 22 / DMS                            |
| DMS <i>classic</i> — SQL Server scenarios   | Retired 15 Mar 2026 → modern DMS (portal / PowerShell / CLI)                    |
| Database Experimentation Assistant (DEA)    | Retired 15 Dec 2024 → open-source SQLWorkload / WorkloadTools                   |
| SQL Data Sync                               | Retires 30 Sep 2027 → use ADF / transactional replication / AG                  |
| DMS → Azure SQL DB                          | Offline only (online available for MI / SQL VM)                                 |
| MI Link source                              | SQL Server 2016 and later (incl. 2022, 2025), Windows Server 2016+, Ent/Std/Dev |
| LRS source                                  | SQL Server 2012+, all editions                                                  |
| Native restore → MI                         | SQL Server 2008+                                                                |
| Transactional replication → MI              | SQL Server 2012+                                                                |
| Transactional replication → SQL DB          | SQL Server 2016–2019 only                                                       |
| Backup-to-URL size                          | 12.8 TB (2016+) / 1 TB                                                          |
| Fabric Migration Assistant                  | Preview; DACPAC ≤ 20 MB; on-prem gateway only                                   |
| SQL Server 2025 (source)                    | GA 18 Nov 2025 — native vector/JSON, improved DAG                               |

## 10 Cross-cloud sources & reverse migration

- **Cross-cloud sources** explicitly supported by DMS / LRS / MI Link / native restore: AWS EC2, AWS RDS for SQL Server, GCP Compute Engine, GCP Cloud SQL for SQL Server.
- **Reverse / exit migration:** SQL DB → SQL Server via BACPAC + scripts (heavy); MI → SQL Server 2022 / 2025 via MI Link reverse failback (trivial — a strategic portability argument).
- **Neutral non-Microsoft targets** (for honest comparison): AWS RDS for SQL Server (+ AWS DMS), GCP Cloud SQL (+ GCP DMS), Tessell, OCI self-managed. Azure differentiators: MI Link, Arc, Fabric Mirroring.

## 11 Third-party alternatives (when they beat the native stack)

| Tool                                     | Better when...                                                                                                                                                                                                                                                                                               |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Striim                                   | Microsoft-recommended <b>online</b> / CDC data-migration vehicle for SQL Server → Azure SQL Database (the online path DMS lacks for SQL DB) and for heterogeneous sources (Sybase / Oracle / DB2 → Azure SQL, paired with SSMA for schema); also real-time CDC to Event Hubs / Synapse / Cosmos in parallel. |
| Dr Migrate / CAST Highlight / UnifyCloud | Portfolio & application/code assessment at scale (wave planning, dependency mapping) — complementary to Azure Migrate.                                                                                                                                                                                       |
| Qlik Replicate                           | Heterogeneous sources (Oracle, DB2, iSeries, SAP HANA) with in-flight transforms — more mature than DMS for Oracle→SQL.                                                                                                                                                                                      |
| Fivetran HVR                             | Broad CDC + observability; multi-target (Snowflake + Fabric).                                                                                                                                                                                                                                                |
| Quest SharePlex                          | SQL Server ↔ Oracle replication; de-Oracle-ization projects.                                                                                                                                                                                                                                                 |
| Carbonite Migrate                        | VM-level always-on with failback when Azure Migrate/ASR can't handle the OS/hypervisor (KVM, Xen, Citrix).                                                                                                                                                                                                   |
| Veeam / Commvault / Cohesity / Rubrik    | Already in place — push backups to Blob, restore on SQL VM (no double licensing).                                                                                                                                                                                                                            |
| Redgate / Liquibase / Flyway             | Schema versioning (GitOps) post-migration — SqlPackage's weak spot.                                                                                                                                                                                                                                          |
| VMware HCX (Broadcom)                    | The zero-downtime "as-is" option for complex VMware clusters incl. SQL FCI → AVS.                                                                                                                                                                                                                            |

## 12 Decision criteria & “when to recommend what”

| Criterion                   | VM         | AVS          | MI              | SQL DB           | Fabric SQL DB   | AKS / Arc |
|-----------------------------|------------|--------------|-----------------|------------------|-----------------|-----------|
| OS access required          | ✓          | ✓            | ✗               | ✗                | ✗               | partial   |
| Cross-DB / DTC transactions | ✓          | ✓            | ✓               | ✗                | ✗               | ✓         |
| FILESTREAM / FileTable      | ✓          | ✓            | ✗               | ✗                | ✗               | ✗         |
| PolyBase                    | ✓          | ✓            | ✗               | ✗                | ✗               | ✓         |
| Service Broker              | ✓          | ✓            | ✓ <sup>1</sup>  | ✗                | ✗               | ✓         |
| SQL CLR / linked servers    | ✓          | ✓            | ✓               | ✗                | ✗               | ✓         |
| SQL Agent                   | ✓          | ✓            | ✓               | ✗ (Elastic Jobs) | ✗               | ✓         |
| Min. downtime achievable    | ~h (DAG)   | ~h (vMotion) | ~min (MI Link)  | min-h            | h               | depends   |
| Managed patch / upgrade     | Auto-patch | ✗            | ✓ Evergreen     | ✓ Evergreen      | ✓ Evergreen     | ✗         |
| Azure Hybrid Benefit        | ✓          | ✓            | ✓               | ✓ (vCore)        | n/a             | ✓         |
| Sovereignty / edge          | ✓          | ✓            | limited regions | limited regions  | limited regions | ✓ (Arc)   |

<sup>1</sup> Azure SQL MI supports Service Broker **within a single instance** only — no cross-instance Service Broker routing.

| Client profile                           | Neutral recommendation           | Why                                     |
|------------------------------------------|----------------------------------|-----------------------------------------|
| Banking / regulated / strong on-prem     | SQL VM + ESU via Arc, or AVS     | OS control, compliance, soft transition |
| Multi-tenant SaaS ISV                    | SQL DB Hyperscale / Elastic Pool | Per-tenant elasticity, cost control     |
| Heavy legacy ERP (SAP, on-prem Dynamics) | SQL MI or SQL VM                 | Instance & SQL Agent compatibility      |
| Modern micro-services app                | SQL DB serverless                | Pay-per-use, auto-pause                 |
| Fabric-native / analytics-first          | SQL DB in Fabric + Mirroring     | OLTP + OneLake unification              |
| Edge / sovereign / multi-cloud           | Arc-enabled SQL MI on local AKS  | Sovereignty + Azure consistency         |
| Short-term data-center exit              | AVS + VMware HCX                 | Zero refactor, vMotion                  |
| Modernize off end-of-support             | SQL VM “as-is” + free ESU        | 3 years free ESU on Azure VM            |

## 13 Field insights — recurring pitfalls

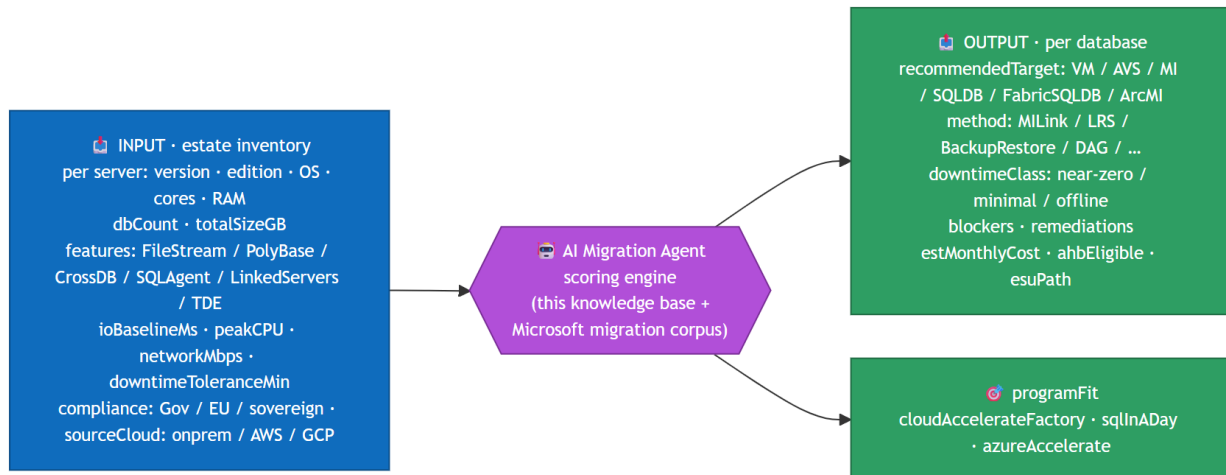
- **TDE certificate:** protect-then-restore order matters — migrate the server cert first, or native restore fails ~80% in with no clear message.
- **Windows logins:** DMS skips them by default; enable the option and grant MI read to Entra ID (Privileged Role Administrator).
- **MI Link ports:** needs 5022 both directions — frequently blocked in banking/industrial networks without a security exception.
- **Other methods need network too:** DMS / LRS / transactional replication require outbound HTTPS (443) to Azure Storage/Blob, SQL 1433 (and 1434/UDP SQL Browser for named instances) — anticipate firewall/NSG blocks in locked-down environments.
- **DAG:** requires AD Domain Services (or workgroup AG + certs) — an infra blocker architects forget.
- **Transactional replication → SQL DB:** SQL 2016–2019 publishers only, with article-type limits — audit before promising.
- **Hyperscale:** the only viable SQL DB choice above 4 TB or with heavy concurrent write I/O.

- **Fabric SQL DB:** Preview for migration, 20 MB DACPAC cap, no Private Link — don't position as enterprise prod yet.
  - **Mirroring ≠ migration:** it's continuous analytics replication; treating it as a one-shot migration is a dangerous shortcut.
  - **Dependency mapping:** ~60% of migrations stumble on undocumented linked servers and SQL Agent jobs — run a dependency map (Azure Migrate or third-party) before committing a target.
  - **SLAs differ:** MI BC 99.99% · SQL DB BC 99.995% (zone-redundant) · SQL DB Hyperscale 99.99% · SQL VM depends on the AG. Align to the app, not a slogan.
  - **Security by design:** SQL DB has a public endpoint by default — recommend Private Endpoint + Entra-only auth + disabled SQL auth from day one.
  - **Land the zone first:** stand up an Azure **landing zone** (CAF — IAM, policy, networking, monitoring, Defender pre-baked) before migrating — Microsoft reports ~4x faster migration/modernization once it's deployed, because each workload move stops being a one-off.
  - **Fabric SQL DB ≠ Azure SQL DB:** not a drop-in replacement — fine-grained security (RLS / OLS / dynamic masking) does not propagate to OneLake; governance must be re-implemented in Fabric. For analytics on operational data, prefer Mirroring (keep OLTP on Azure SQL, mirror into Fabric, no ETL) over a hard cutover.
  - **Bootstrap via Arc first:** connect the estate to Azure Arc *before* migrating — free weekly continuous assessment, blocker detection, ESU and PAYG (OpEx) licensing while you plan, with zero workload change (§3 row 8).
- 

## 14 FY27 SQL Motion context & AI Migration Agent

This document is the knowledge base behind the **FY27 EMEA EPS — Data Motion “SQL in a Day”**. Two notes:

- **Storyline correction.** The deck line “Azure Migrate, DMA, DMS, SSMA, Cloud Accelerate Factory” must drop DMA (retired) → use “Azure Migrate, SSMS 22 + Arc-based assessment, DMS, SSMA, Cloud Accelerate Factory”. A 4th target pillar — SQL Server enabled by Azure Arc — should appear alongside VM / MI / SQL DB.
- **AI Migration Agent — I/O contract.** The afternoon agent scores the best path on the customer estate. Suggested flow so it's usable by both humans and automation:



**Microsoft programs to attach:** Cloud Accelerate Factory (zero-cost delivery), Azure Accelerate / FastTrack, AHB + ESU via Arc — detailed in §15.

## 15 Commercial levers & funding programs (FY27)

Two money levers to combine on every deal: **(A) Microsoft-funded engagement** money (assess / pilot / migrate) and **(B) commercial / licensing** levers that durably cut TCO. **Amounts, ratios and program names are partner-confidential and drift each fiscal year (FY27 starts 1 Jul 2026) — re-validate live in Partner Center / the MCI guide before quoting.**

### 15.1 Commercial / licensing levers (permanent TCO reducers)

| Lever                        | What it saves                                 | Mechanics                                                                                                                                                                  |
|------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Azure Hybrid Benefit (AHB)   | 30%+ on Azure SQL DB / MI; compute-only on VM | License + Software Assurance reallocated to Azure; 1 Enterprise core = 4 GP vCores; 180-day dual-use during migration; portal toggle. Not for the DTU model or serverless. |
| Free ESU on Azure            | Removes on-prem ESU cost                      | Azure VMs / AVS: ESU free & automatic; Azure SQL PaaS: ESU charges stop; on-prem / multicloud via Arc: ESU paid; non-prod free when prod runs ESU via Arc.                 |
| PAYG licensing via Arc       | Turns the SQL license into OpEx               | Billed only when SQL runs; CALs included; can sit inside a MACC. Requires active SA or PAYG enabled.                                                                       |
| Free Azure SQL offers        | Zero-cost POC / pilots                        | MI free 12 months; SQL DB free for the subscription lifetime (serverless GP).                                                                                              |
| Reservations / Savings Plans | 1- / 3-yr commitment discount                 | Stacks with AHB. Partner Earned Credit (15%) does not apply to reservations.                                                                                               |
| Savings plan for databases   | up to 35% on Azure SQL (DB / MI)              | 1- or 3-year hourly compute commitment; auto-applies across participating database services up to the commitment; stacks with AHB.                                         |

## 15.2 Microsoft-funded engagement programs

| Program                                                                                                                            | Funds                                                                          | Access                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Azure Accelerate (FY26 umbrella = <i>Azure Migrate &amp; Modernize</i> + <i>Azure Innovate</i> + <i>Cloud Accelerate Factory</i> ) | Assessments, POC, pilots, deployments, Azure credits, skilling                 | Self-serve — Partner Center nomination → POE → paid ≤ 45 days                   |
| Azure Accelerate for Databases                                                                                                     | SQL / data-estate modernization                                                | Same workflow (database scenario)                                               |
| Cloud Accelerate Factory                                                                                                           | Zero-cost Microsoft expert deployment                                          | Via Azure Accelerate in Partner Center                                          |
| FastTrack for Azure                                                                                                                | Free Microsoft engineering guidance (not cash)                                 | Account team / program alias                                                    |
| ECIF (End-Customer Investment Funds)                                                                                               | Assessments, POCs, migrations, training                                        | Field-led via PDM / AE — not self-serve                                         |
| Azure Frontier Offer (ECIF + ACO, dual-run offset)                                                                                 | Services + dual-run during migration; competitive Oracle / legacy displacement | Field-led via PDM; typically ≥ 50% ACR from Fabric / first-party Databases / AI |

## 15.3 Partner mechanics

- **Attribution first** (funding prerequisite): link the customer via PAL (Partner Admin Link) or DPOR → unlocks the 15% Partner Earned Credit on managed Azure consumption.
- **Routes:** *Self-serve* (Partner Center) — Azure Accelerate, Databases, Cloud Accelerate Factory · *Field-led* (engage the PDM early) — ECIF, Azure Frontier Offer · *Customer self-serve* (Azure portal) — AHB, ESU, free offers, reservations.
- **Typical sequence:** Azure Migrate discovery + TCO → nominate assess/pilot in Azure Accelerate (Databases) → if competitive or ≥ 50% data ACR, engage PDM for the Azure Frontier Offer → pilot on free SQL DB / MI → CAF for zero-cost deploy → POE → lock run-rate with reservations.

## 16 Sources (Microsoft Learn)

**Overviews & taxonomy** - Database Migration hub — <https://learn.microsoft.com/en-us/data-migration/> - SQL Server → Azure SQL Database — <https://learn.microsoft.com/en-us/data-migration/sql-server/database/overview> - SQL Server → Azure SQL Managed Instance — <https://learn.microsoft.com/en-us/data-migration/sql-server/managed-instance/overview> - SQL Server → SQL Server on Azure VM — <https://learn.microsoft.com/en-us/data-migration/sql-server/virtual-machines/overview> - Azure SQL feature comparison — <https://learn.microsoft.com/en-us/azure/azure-sql/database/features-comparison>

**Control planes / tools** - Azure Migrate (SQL assessment) — <https://learn.microsoft.com/en-us/azure/migrate/how-to-create-azure-sql-assessment> - SQL Server migration in Azure Arc — <https://learn.microsoft.com/en-us/sql/sql-server/azure-arc/migration-overview> - SSMS (download/overview) — <https://learn.microsoft.com/en-us/sql/ssms/sql-server-management-studio-ssms> - Azure Database Migration Service — <https://learn.microsoft.com/en-us/azure/dms/dms-overview> - DMS supported scenarios (offline/online per target) — <https://learn.microsoft.com/en-us/azure/dms/resource-scenario-status> - Az.DataMigration PowerShell — <https://learn.microsoft.com/en-us/powershell/module/az.datamigration/> - SSMA — <https://learn.microsoft.com/en-us/sql/ssma/sql-server-migration-assistant> - Database Experimentation Assistant (DEA) — <https://learn.microsoft.com/en-us/previous-versions/sql/dea/database->

[experimentation-assistant-overview](https://learn.microsoft.com/en-us/sql/tools/whats-happening-azure-data-studio) - What's happening with Azure Data Studio — <https://learn.microsoft.com/en-us/sql/tools/whats-happening-azure-data-studio>

**Methods** - Managed Instance link — <https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/managed-instance-link-feature-overview> - Log Replay Service — <https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/log-replay-service-migrate> - Native backup & restore (MI) — <https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/restore-sample-database-quickstart> - Transactional replication (SQL DB) — <https://learn.microsoft.com/en-us/azure/azure-sql/database/replication-to-sql-database> - Import/Export · BACPAC — <https://learn.microsoft.com/en-us/azure/azure-sql/database/database-import-bcp-utility> - bcp utility — <https://learn.microsoft.com/en-us/sql/tools/bcp-utility> - Smart Bulk Copy — <https://github.com/Azure-Samples/smartbulkcopy> - Backup to URL — <https://learn.microsoft.com/en-us/sql/relational-databases/backup-restore/sql-server-backup-to-url> - Availability group → Azure VM — <https://learn.microsoft.com/en-us/data-migration/sql-server/virtual-machines/availability-group-migrate> - Log shipping — <https://learn.microsoft.com/en-us/sql/database-engine/log-shipping/about-log-shipping-sql-server> - Elastic Jobs (SQL DB) — <https://learn.microsoft.com/en-us/azure/azure-sql/database/elastic-jobs-overview>

**Targets — containers, Fabric, AVS, Arc** - Create Arc-enabled SQL Managed Instance — <https://learn.microsoft.com/en-us/azure/azure-arc/data/create-sql-managed-instance> - Migrate to Arc-enabled SQL MI — <https://learn.microsoft.com/en-us/azure/azure-arc/data/migrate-to-managed-instance> - SQL Server on Kubernetes / AKS — <https://learn.microsoft.com/en-us/sql/linux/quickstart-sql-server-containers-kubernetes> - SQL Server in a Docker container — <https://learn.microsoft.com/en-us/sql/linux/quickstart-install-connect-docker> - Fabric Migration Assistant for SQL database (Preview) — <https://learn.microsoft.com/en-us/fabric/database/sql/migration-assistant> - Migrate to Fabric SQL DB via DACPAC — <https://learn.microsoft.com/en-us/fabric/database/sql/migrate-with-migration-assistant-using-dacpac> - Fabric Mirroring for SQL Server — <https://learn.microsoft.com/en-us/fabric/mirroring/sql-server> - Limitations for SQL database in Fabric — <https://learn.microsoft.com/en-us/fabric/database/sql/limitations> - Fabric — choose a data store (decision guide) — <https://learn.microsoft.com/en-us/fabric/fundamentals/decision-guide-data-store> - Azure VMware Solution — <https://learn.microsoft.com/en-us/azure/azure-vmware/introduction> - SQL MI — Next-gen General Purpose — <https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/service-tiers-next-gen-general-purpose-use> - Azure SQL DB Hyperscale — <https://learn.microsoft.com/en-us/azure/azure-sql/database/service-tier-hyperscale>

**Data movement & VM-level** - Azure Data Box / Data Box Heavy — <https://learn.microsoft.com/en-us/azure/databox/data-box-overview> - ExpressRoute — <https://learn.microsoft.com/en-us/azure/expressroute/expressroute-introduction> - AzCopy — <https://learn.microsoft.com/en-us/azure/storage/common/storage-use-azcopy-v10> - Azure Storage Mover — <https://learn.microsoft.com/en-us/azure/storage-mover/service-overview> - Azure Site Recovery — <https://learn.microsoft.com/en-us/azure/site-recovery/site-recovery-overview> - Azure-SSIS Integration Runtime — <https://learn.microsoft.com/en-us/azure/data-factory/create-azure-ssis-integration-runtime> - SQL Data Sync (retiring 30 Sep 2027) — <https://learn.microsoft.com/en-us/azure/azure-sql/database/sql-data-sync-data-sql-server-sql-database?view=azuresql>

**Licensing / programs / lifecycle** - Azure Hybrid Benefit (Azure SQL) — <https://learn.microsoft.com/en-us/azure/azure-sql/azure-hybrid-benefit> - SQL Server ESU enabled by Azure Arc — <https://learn.microsoft.com/en-us/sql/sql-server/azure-arc/extended-security-updates> - Move SQL Server license to PAYG (Arc) — <https://learn.microsoft.com/en-us/sql/sql-server/azure-arc/manage-pay-as-you-go-transition> - SQL Server end-of-support / ESU — <https://learn.microsoft.com/en-us/sql/sql-server/end-of-support/sql-server-extended-security-updates> - SQL Server 2016 lifecycle — <https://learn.microsoft.com/en-us/lifecycle/products/sql-server-2016> - Azure SQL MI free offer — <https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/free-offer> - Azure SQL DB free offer — <https://learn.microsoft.com/en-us/azure/azure-sql/database/free-offer> - Azure Accelerate (partner programs) — <https://partner.microsoft.com/en-us/partnership/azure-offerings> - Tool consolidation / retirement (blog) — <https://www.microsoft.com/en-us/sql-server/blog/2024/09/12/modernize-your-database-with-the-consolidation-and-retirement-of-azure-database-migration-tools/> - Cloud Adoption Framework — migrate — <https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/migrate/> - Azure landing zones (CAF) — <https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/> - Savings plan for databases — <https://learn.microsoft.com/en-us/azure/cost-management-billing/savings-plan/savings-plan-overview> - Modernize your databases (hub) — <https://aka.ms/modernizedatabases>

*Links last verified: July 2026. Microsoft migration guides moved from .../azure-sql/migration-guides/... to .../data-migration/sql-server/... (redirects in place). Items marked **Preview** are subject to change.*

## 17 Document version & changelog

Current version: **v1.2** (2026-07-03).

### Version history (current: v1.2)

| Version | Date       | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1.2    | 2026-07-03 | Corrected two moved Microsoft Learn links: Smart Bulk Copy (now the Azure-Samples GitHub repo) and Migrate to Arc-enabled SQL MI (renamed to azure-arc/data/migrate-to-managed-instance). Added the weekly link + news freshness automation (GitHub Action).                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| v1.1    | 2026-07-03 | Azure SQL MI <b>Next-gen General Purpose</b> re-classified <b>preview</b> → <b>GA</b> (GA since Nov 2025). Dates refreshed to July 2026 (verification note, lifecycle status column, sources footer). All ~45 Microsoft Learn links re-verified. Re-confirmed still-current: Fabric Migration Assistant remains Preview (DACPAC ≤ 20 MB, on-prem gateway only, no Private Link); SSMS 22 Azure SQL assessment still roadmap ~Q3 CY2026; SQL Server 2016 ESU free on Azure from 14 Jul 2026; MI Link GA with SQL Server 2025 support + reverse fallback to 2022/2025.                                                                                                                                                             |
| v1.0    | 2026-06    | Initial published knowledge base: 8 target families (SQL VM, AVS, SQL MI, SQL DB, Fabric SQL DB, containers AKS/ARO/ACI/ACA, Arc-enabled SQL MI, Arc in-place); targets/control-planes/methods taxonomy; per-target method tables (min source · downtime · constraints); 2025–2026 tooling reset (DMA, ADS, DMS classic, SQL Data Sync retirements); tooling-by-source matrix (incl. STRIIM online to Azure SQL DB); downtime strategy; summary & decision matrices; ancillary components; cross-cloud & reverse migration; third-party alternatives; field insights; commercial & funding levers (AHB / ESU / PAYG · Savings plan for databases · Azure Accelerate); FY27 SQL Motion context & AI Migration Agent I/O contract. |

Versioning: **MAJOR.MINOR** — MINOR bumps for fact/link refresh and additive content; MAJOR bumps for structural rewrites or scope changes.

## 18 Appendix — Using the sql-migration-advisor Copilot CLI skill

This knowledge base also powers sql-migration-advisor, a GitHub Copilot CLI skill that turns the inventory above into a guided recommendation. Ask Copilot “*migrate a SQL Server environment to Azure*”: it runs a short ~10-question interview (scope, source location & version, primary driver, instance-level feature dependencies, largest DB size, downtime tolerance, network & ports, compliance, ancillary services), then returns a grounded recommendation card — target, method, downtime class, blockers + remediations, cost levers, and the right Microsoft program — and never recommends retired tooling (DMA, Azure Data Studio extension, DMS classic).

### 18.1 Install (once)

macOS / Linux:

```
git clone https://github.com/fredgis/sql-migration-advisor.git
↪ ~/.copilot/skills/sql-migration-advisor
```

Windows (PowerShell):

```
git clone https://github.com/fredgis/sql-migration-advisor.git
↪ "$env:USERPROFILE\.copilot\skills\sql-migration-advisor"
```

Restart Copilot CLI, run /skills to confirm sql-migration-advisor is listed, then ask your question in natural language (English or French).

### 18.2 Example output

The card below is a real recommendation for an estate of ~100 on-premises SQL Server 2012 databases (limited WAN, exiting out-of-support):

Verdict - estate of ~100 databases (on-prem SQL 2012 + PaaS)  
Azure SQL Managed Instance (General Purpose) via LRS in seed-then-sync · downtime minimal · grounded in the FY27 SQL knowledge base.

Fully-managed PaaS + “not sure” dependencies + exiting out-of-support SQL 2012: SQL MI is the default managed lift-and-shift. It absorbs your 150 GB+ TB databases in General Purpose, handles any undocumented dependencies (linked servers, SQL Agent), and MI Link is excluded because the 2012 source is below the required 2016+ → the method is LRS.

**Estate strategy (100 databases)**

1. Azure Migrate: agentless discovery, dependency mapping, costed business case, and MI right-sizing.
2. **Az.DataMigration** (CLI/PowerShell): industrializes LRS beyond ~50 databases - the only viable path at this scale.
3. Data Box for the initial seed (limited WAN), then LRS catches up the log delta before each cutover.

**At a glance**

|                      | Recommendation                                                               |
|----------------------|------------------------------------------------------------------------------|
| Target               | Azure SQL Managed Instance - General Purpose (size 150 GB+ TB, managed PaaS) |
| Method               | Log Replay Service (LRS) - 2012 source OK; MI Link excluded (requires 2016+) |
| Downtime             | Minimal (LRS replays full/diff/log, short cutover)                           |
| Assess / orchestrate | Azure Migrate → Az.DataMigration at scale → SSMS 22 / Arc                    |

**Blockers & fixes**

- Limited WAN + multi-TB + Data Box for the initial seed, then LRS sync (seed-then-sync pattern); otherwise timeouts are guaranteed.
- Out-of-support SQL 2012 source → migrate fast; restoring via LRS removes the need for ESU.
- “Not sure” dependencies → run an Azure Migrate dependency map before locking the target <40% of migrations stumble on forgotten linked servers / SQL Agent jobs).
- Logins / users → script and recreate (not migrated by default).
- EU data residency → MI region within the EU perimeter from creation.
- Network → validate outbound nls (9140) and nls across all 100 sources.

**Cost & program (your question)**

- Azure Hybrid Benefit (AHB) - eligible on SQL MI (Cores) - reuse your SQL licenses with SA, up to ~80% savings combined with 1-3 year reservations.
- ESU = the real “stay vs migrate” lever: staying on SQL 2012 on-prem forces paid ESU via Azure Arc; migrating to MI (evergreen) eliminates that cost entirely.
- Consolidation: 100 databases → group onto a few MI instances (instance pools / shared vCores) instead of 100 isolated targets - a major cost lever.
- GP vs RC: stay General Purpose (pay-per-minute storage) unless you need local latency + Business Critical.
- Sizing: baseline Perform > 7 days + ~20% headroom, never average CPU - over-sizing across 100 databases is expensive.
- Programs: Cloud Accelerate Factory (zero-cost delivery), SQL in a Day, Azure Accelerate / FastTrack - attach to fund the delivery.

⚠ Biggest risk: network throughput over a limited WAN with 100 databases. Don't compute **size + bandwidth** - test AzCopy + one real full backup, seed via Data Box, and plan a rollback window. Validate target performance with DEA (capture + replay) on sensitive databases.

🔗 LRS · Az.DataMigration · Azure Migrate - business case · AHB - LRS: planned offline, no read-only target.

Would you like a per-database-group table, a scaled LRS runbook, or a summary slide for the estate?